Rotary Wing Aerodynamics AE 667

Dhwanil Shukla



The course will…

- Teach you a ton about rotors
- Force you to do open-ended assignments
- Give you some programming practice
- Probably make you better engineers overall

(no promises!)

Course Outline

1. Introduction
2. Rotor Hover Performance Prediction Methods
3. Vertical Descent of Rotors
4. Rotors in Forward Flight
5. Vehicle Performance
6. Advanced Topics

Learning Outcomes

By the end of the course, we should be able to:

1. Analyse & design rotors/propellers for performance in hover/climb using Momentum theory and Blade Element Momentum theory

2. Predict basic helicopter blade and rotor dynamics in a steady level forward flight

3. Find pilot inputs necessary to achieve trim conditions in a steady level forward flight

4. Estimate helicopter performance based on design parameters

5. Follow current research in the field of rotary wing aerodynamics

Reference Books

- Venkatesan, C. (2014). *Fundamentals of helicopter dynamics*. CRC Press.

- Johnson, W. (2012). *Helicopter theory.* Courier Corporation.

- Leishman, G. J. (2006). *Principles of helicopter aerodynamics.* Cambridge university press

[Lecture notes will also be made available after the classes]

Evaluation, Grading, Attendance

- Exams: 2 (10% each)
- Assignments: 3 ***(read: mini projects)*** (20% each)
- Literature Survey Project (group effort) (20%)

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* Grades to be curved based on the class performance

* Fail Grade for overall score less than 40%

* Audit Criteria: Score at least 40% in the course,
  Satisfy Attendance Policy

* Attendance Policy: No performance stress < 70% attendance!

Discussions vs Plagiarism

- **Discussions** on assignment problems **encouraged *well before*** the submission deadlines.
- **Sharing ≠ Caring**
  Sharing assignment reports and codes not a way to help!
  Explain concepts and help your friend do the job by self…
- **100% Discount Offer!!**
  In marks for ALL involved in Copying/Plagiarising assignments.
  [*Yes, we do have tools to detect plagiarism in computer codes as well*]
- **Repeated** 100% discount seekers to get the entire course free or cost.

Assignment Theme:

Indigenous High Speed Troop Transport Compound Helicopter

- 400 km/h top speed
- 5000 m service ceiling
- 500 km range
- ALH Druv platform
- 2 pilots + 8 passengers





Introducing Rest of the Team

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Don’t hesitate to use the assistance you are provided with to succeed!

Typical Lecture Structure…

1. Join precisely on time..
2. Discuss some interesting engineering fact/news – 5 min
   On nomination basis, For each lecture
3. Recall the glorious past (Recap & attendance) – 2-3 min
4. Proceed in the course…
5. [Interruptions for relevant questions and discussions highly encouraged!]
6. Reflections (lucky draw) – 5 min

Introduction & History

First Vertical Flight



Paul Cornu (1907)
The first flying machine to have risen from the ground using rotor blades instead of wings.
[Wright brothers' first flight in Dec 1903]

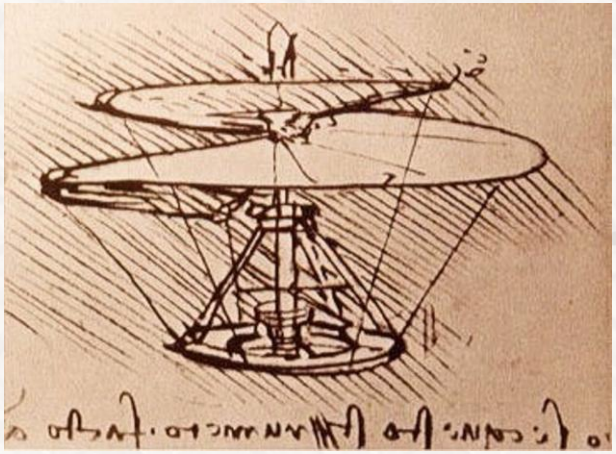
First Helicopter

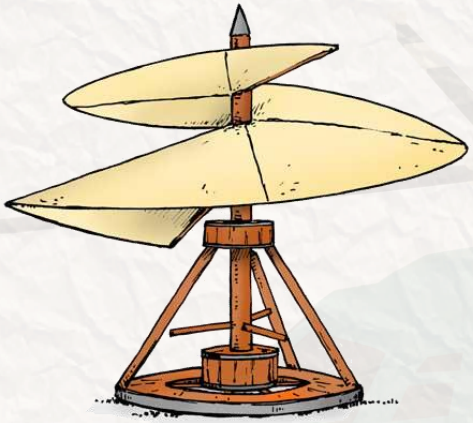


TESTING THE CONTROLS. Mr. Igor Sikorsky makes the first ascent on his new helicopter. Note that the machine is "tethered." Subsequently several free flights were made.

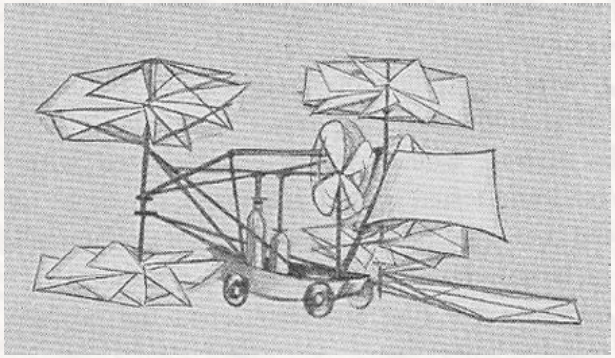
Igor Sikorsky, 14th September 1939

Early Ideas





Leonardo da Vinci
(15th century)



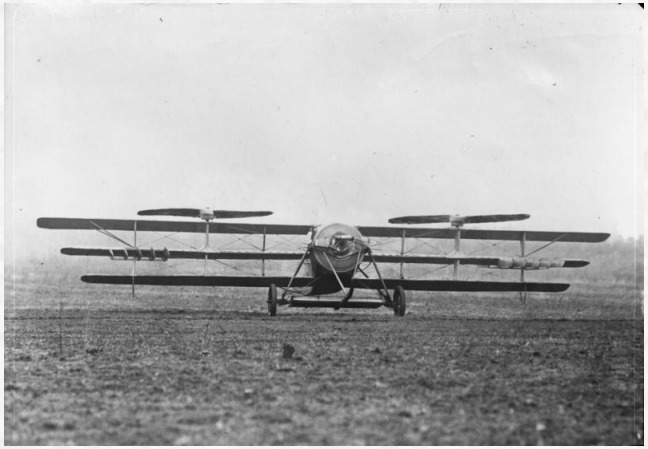
Sir George Cayley (1796)

Early Ideas



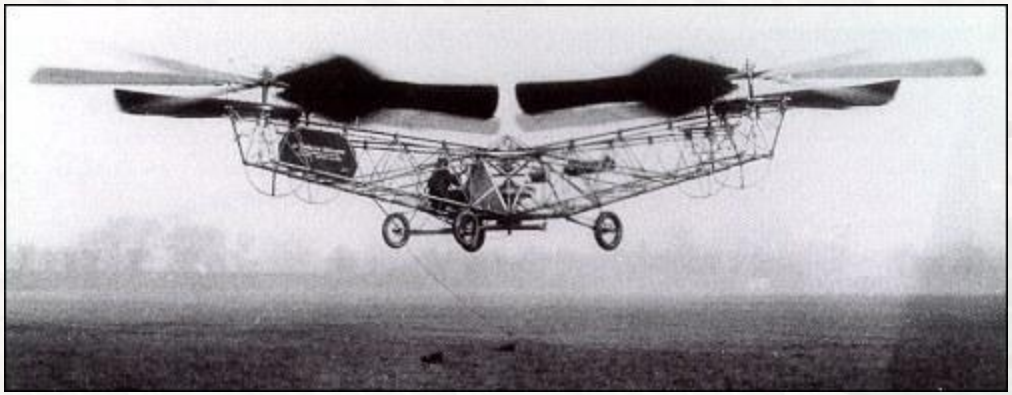


Emile & Henry Berliner (1908 - 1929)
- Coaxial Helicopter
- Tilt Rotor

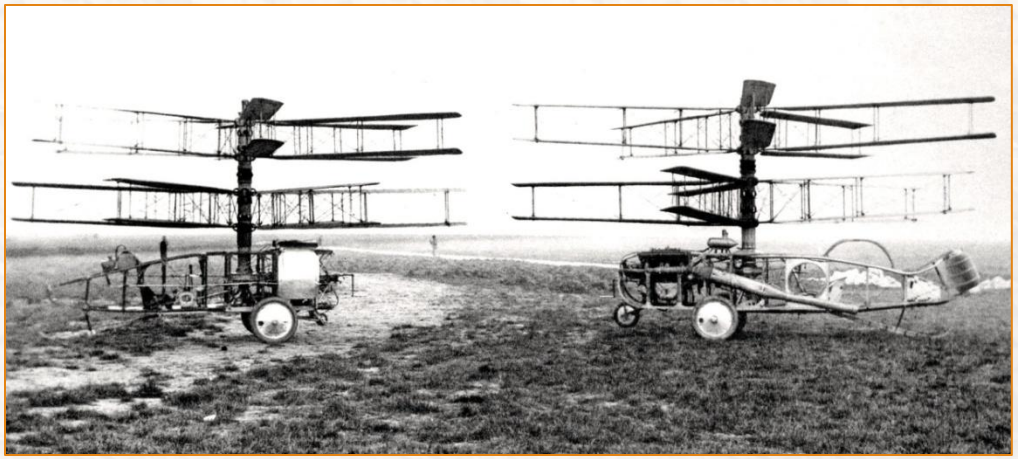




Early Ideas

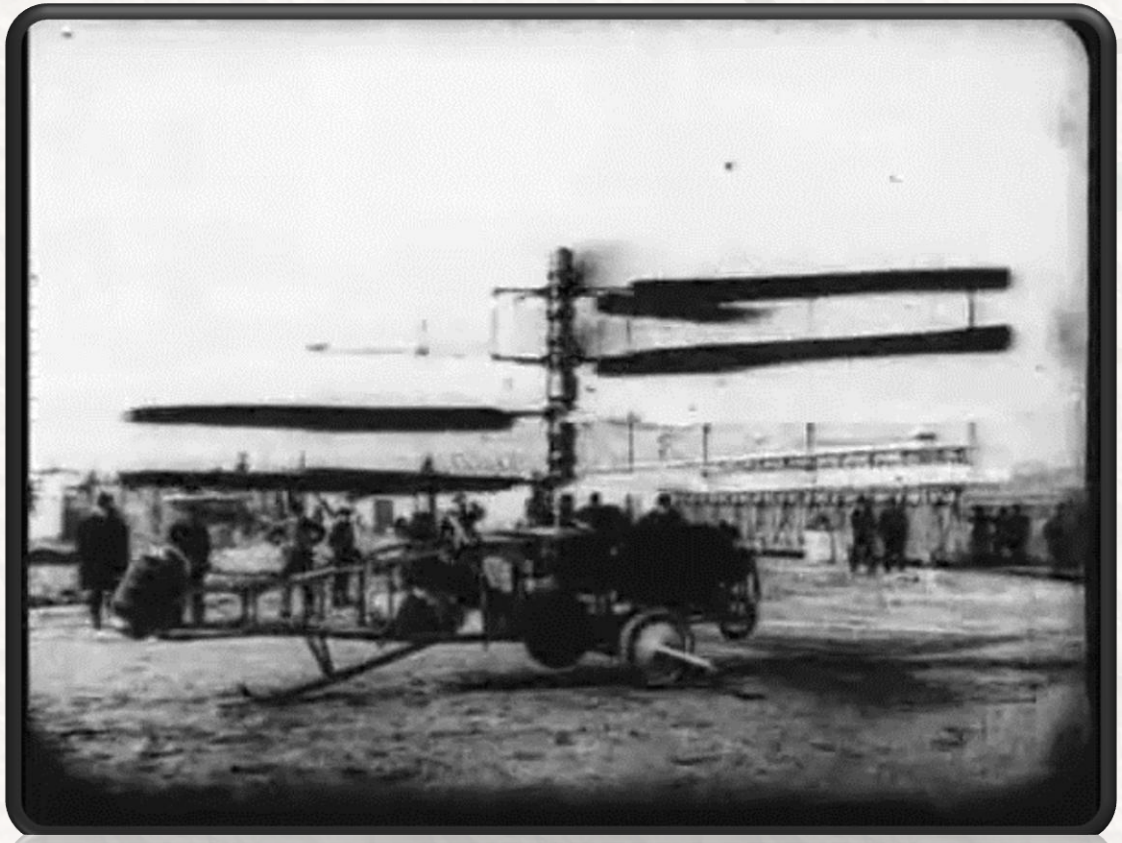


De Bothezat's Quadcopter (1921)

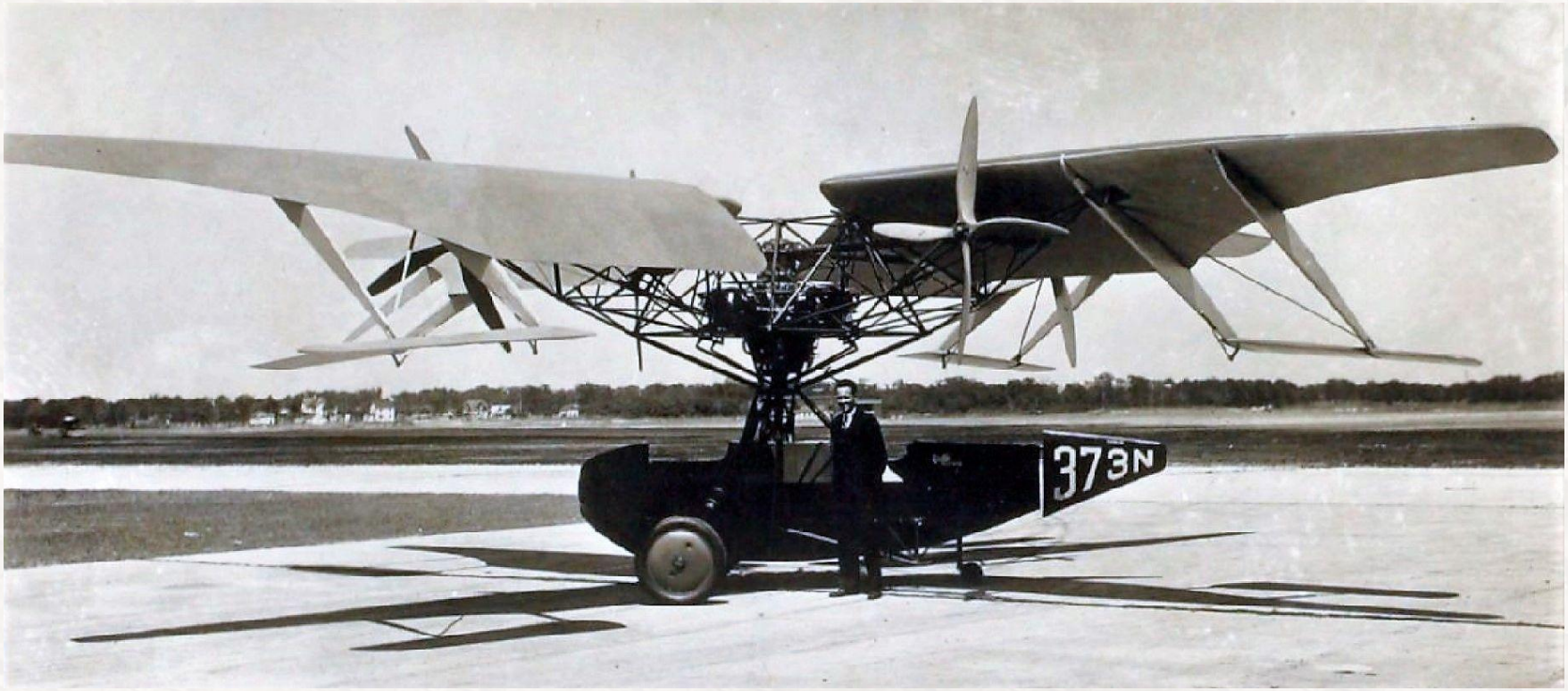


Pescara Coaxial Helicopter (1925)

Early Ideas



Early Ideas



Curtiss-Bleeker helicopter (1930)

Present day helicopters



Conventional



Coaxial

Present day helicopters



V-22 Osprey
Tiltrotor



CH-47 Chinook
Tandem

World’s largest



Mil Mi-26



Mil V-12

India’s Indigenous Helicopters



Dhruv
(Advanced Light Helicopter)



Prachand
(Light Combat Helicopter)

Problems of Forward Flight

